SelectiveNet：集成拒绝选项的深度神经网络

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摘要

我们考虑深度神经网络中的选择性预测问题（也称拒绝选项），提出SelectiveNet，一种集成拒绝选项的深度神经架构。现有拒绝机制主要对预训练网络的预测置信度设定阈值。与之相反，SelectiveNet以端到端方式，同时优化分类（或回归）和拒绝。结果是一个针对被覆盖区域优化的深度神经网络。实验显示，在多个知名分类与回归数据集上，风险—覆盖率权衡持续改善，从而在深度选择性分类中取得新的最佳结果。

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1 引言

在自动驾驶、医疗诊断、家庭机器人等许多关键任务的机器学习应用中，检测与控制学习过程中的统计不确定性至关重要。选择性预测旨在学习“知道自己不知道什么”的预测模型。当模型对预测不够有信心时，允许它弃权。

拒绝选项的思想早在60多年前就由Chow（1957）研究过。许多假设类和学习算法都已考虑拒绝机制，主要由两种相关思想驱动。第一种是基于成本的模型：使选择性模型的损失最小，损失函数记录用户指定的弃权成本。另一种以预测置信度代理函数为基础；典型例子包括函数间隔、到最近邻的距离等。

本文关注深度神经网络（DNN）的选择性模型。最近的相关工作是Geifman与El-Yaniv（2017），展示如何针对给定的、已训练好的模型，使用任意确定性估计函数构造经过概率校准的选择性分类器。有趣的是，简单的Softmax响应（SR）估计器是已训练网络的最佳确定性代理之一，实践者也经常使用。

在用户指定目标覆盖率的弃权约束框架下，我们端到端优化一个集成拒绝选项的神经网络，使其针对要求的覆盖率水平达到最优。该方法产生有效的选择性模型，持续改善已知技术的效果。

从学习理论角度，不难论证：为每个覆盖率水平学习专门的拒绝模型，可能优于根据预训练网络提取的置信度进行拒绝。下面的小故事说明专门训练的优势。Bob和Alice即将参加机器学习课程考试。考试有10道题，每道对应一个课程主题，但学生只需选择恰好5道。Alice只复习最简单的5个主题，而Bob的妈妈要求他学习全部10个主题。你认为谁会得到更高的分数?

本文贡献为:

- 一种使用内点优化方法变体（Potra与Wright，2000）、针对指定覆盖率水平优化的选择性损失函数。
- SelectiveNet：端到端学习选择性分类模型的三头网络。
- 用于回归的SelectiveNet：为MC-dropout、集成技术等高成本方法提供首个替代方案。
- 在四个分类与回归数据集上展示经验优势，相较MC-dropout和SR有显著提升。

2 选择性预测问题的形式化

我们考虑分类、回归等任意选择性预测任务。监督预测任务形式如下。令$\mathcal{X}$为任意特征空间，$\mathcal{Y}$为标签空间。例如，$\mathcal{X}$可以是图像，分类时$\mathcal{Y}$可为类别标签，回归时可为边界框坐标。令$P(X,Y)$为$\mathcal{X}\times\mathcal{Y}$上的分布。模型$f:\mathcal{X}\to\mathcal{Y}$称为预测函数，其关于$P$的真实风险为

$$R(f)\triangleq\mathbb{E}_{P(X,Y)}[\ell(f(x),y)],$$

其中$\ell:\mathcal{Y}\times\mathcal{Y}\to\mathbb{R}_+$是给定损失，如0/1损失或平方损失。给定从$P(X,Y)$独立同分布采样的标注集$S_m=\{(x_i,y_i)\}_{i=1}^m\subseteq(\mathcal{X}\times\mathcal{Y})^m$，分类器$f$的经验风险为

$$\hat{r}(f\mid S_m)\triangleq\frac{1}{m}\sum_{i=1}^m\ell(f(x_i),y_i).$$

选择性模型（El-Yaniv与Wiener，2010）是二元组(f,g)。其中f为预测函数，$g:\mathcal{X}\to\{0,1\}$为选择函数，作为f的二元资格判定器:

$$(f,g)(x)\triangleq\begin{cases}f(x), & g(x)=1;\\ \text{不知道}, & g(x)=0.\end{cases}$$

因此，当且仅当$g(x)=0$时，模型在x处弃权。也可考虑软选择函数$g:\mathcal{X}\to[0,1]$，以概率方式或相对于阈值确定性地作出决策。

选择性模型的性能由覆盖率和风险量化。覆盖率定义为

$$\phi(g) \triangleq \mathbb{E}_P[g(x)],$$

即$\mathcal{X}$中未被拒绝区域的概率质量。(f, g)的选择性风险为

$$R(f, g) \triangleq \frac{\mathbb{E}_P[\ell(f(x), y)g(x)]}{\phi(g)}.$$

显然，风险可与覆盖率权衡。模型的完整性能可用风险—覆盖率曲线刻画，即风险作为覆盖率的函数（El-Yaniv与Wiener，2010；Wiener与El-Yaniv，2012）。真实选择性风险和覆盖率对应的经验量可在任意标注集S_m上计算：

$$\hat{r}(f, g \mid S_m) \triangleq \frac{\frac{1}{m}\sum_{i=1}^{m} \ell(f(x_i), y_i)g(x_i)}{\hat{\phi}(g \mid S_m)}, \qquad \hat{\phi}(g \mid S_m) \triangleq \frac{1}{m}\sum_{i=1}^{m} g(x_i).$$

最优选择性模型有两种定义方式：在覆盖率约束下优化选择性风险，或反过来。本文关注前者。给定覆盖率$0 < c \leq 1$和假设类Θ，最优模型为

$$\theta^* = \arg\min_{\theta \in \Theta} R(f_\theta, g_\theta) \quad \text{使得} \quad \phi(g_\theta) \geq c. \tag{1}$$

本文中Θ为f和g的给定深度网络架构的参数集合。借助Geifman与El-Yaniv（2017）的方法，可将上述任意控制覆盖率、优化风险的模型，转为控制风险、优化覆盖率的模型。

3 相关工作

拒绝选项的文献非常丰富，主要关注SVM、最近邻、提升等假设类与学习算法（Fumera与Roli，2002；Hellman，1970；Cortes等，2016；El-Yaniv与Wiener，2015）。在神经网络（NN）选择性预测中，一种直接有效的方法是：从已针对所有样本完成预测优化的预训练网络中提取置信分数，并对其设阈值（Cordella等，1995；De Stefano等，2000；Wiener与El-Yaniv，2012）。Geifman与El-Yaniv（2017）将其扩展至DNN，说明如何基于任意置信分数函数构造选择性分类器，以及如何对测试时的选择性风险计算高置信保证。

该工作讨论两种从NN提取置信分数的方法。第一种是Softmax响应（SR），定义为分类架构softmax层的最大激活值。第二种是Gal与Ghahramani（2016）的蒙特卡洛dropout（MC-dropout），通过在网络中施加dropout、进行多次前向传播的统计量估计预测置信度。据我们所知，MC-dropout是从单个网络提取置信分数的唯一非贝叶斯技术。不幸的是，有效MC-dropout每次预测的成本可能达数百次前向迭代。

译注：“唯一非贝叶斯技术”为原文说法；所引MC-dropout论文使用贝叶斯近似解释，且本段也讨论了SR。因此不宜把这一原文措辞当作严谨的当前方法分类结论。

另一类可用于选择性分类的置信分数函数基于多个模型的集成统计量（Lakshminarayanan等，2017；Geifman等，2018）。本文关注单个分类器，同时指出：扩展至选择性分类器集成很直接，而且可能改善结果。

4 SelectiveNet

本节首先定义SelectiveNet，一种支持端到端优化选择性模型的深度神经架构。同时提出适当的损失函数与优化过程，可针对任意目标覆盖率训练。

4.1 架构

SelectiveNet是选择性模型(f, g)，在同一DNN中同时优化$f(x)$与$g(x)$。图1为示意图。输入由主干模块处理，主干由若干DNN层或子模块组成，最后一层称为表示层。主干可采用适用于任务的任意架构，如卷积、全连接或循环架构。

SelectiveNet有预测、选择和辅助预测三个输出头。选择头实现$g(x)$，预测头实现$f(x)$，辅助预测头记为$h(x)$，通过预测相关任务，丰富或强化主干中相关特征的构建。辅助头只在训练时使用，其作用将在定义损失后进一步说明。各头架构可随任务类型与复杂度变化，但选择头$g(x)$的最后一层始终是一个sigmoid激活神经元。$f(x)$的末层依应用而定，可以是softmax（分类）或线性（回归）。推理时，样本x输入SelectiveNet，当且仅当$g(x) \geq 0.5$时输出$f(x)$；否则拒绝预测x的标签。

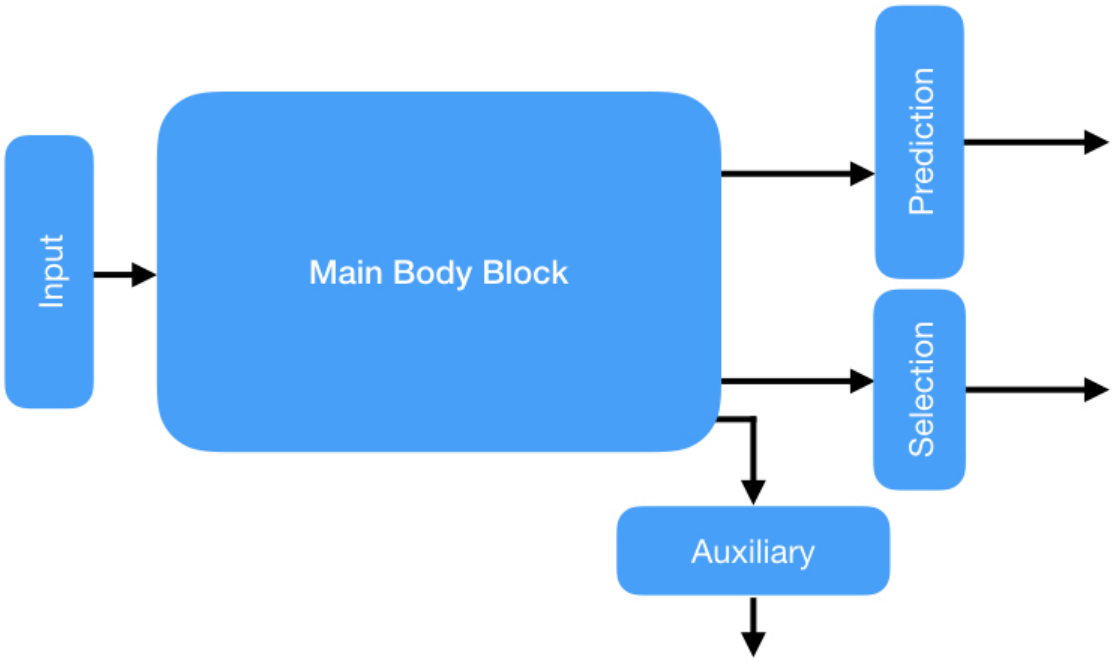


图 1: SelectiveNet架构示意。Input为输入，Main Body Block为主干模块，Prediction为预测，Selection为选择，Auxiliary为辅助头。保留原图英文标签。

4.2 优化

选择性预测目标见式（1）。为施加覆盖率约束，我们采用著名内点法（IPM；Potra与Wright, 2000）的变体，得到在S_m样本上平均的无约束目标：

$$\mathcal{L}_{(f,g)} \triangleq \hat{r}_\ell(f, g \mid S_m) + \lambda\Psi(c - \hat{\phi}(g \mid S_m)), \qquad \Psi(a) \triangleq \max(0, a)^2. \tag{2}$$

c是目标覆盖率，λ控制约束的相对重要性，Ψ是二次惩罚函数。λ的选择见第6节。我们训练辅助头h完成与f相同的预测任务（分类或回归），采用标准损失

$$\mathcal{L}_h = \hat{r}(h \mid S_m) = \frac{1}{m}\sum_{i=1}^{m} \ell(h(x_i), y_i),$$

它不考虑覆盖率。用选择性损失与辅助损失的凸组合优化SelectiveNet，总目标为

$$\mathcal{L} = \alpha\mathcal{L}_{(f,g)} + (1 - \alpha)\mathcal{L}_h.$$

辅助头h对优化至关重要。没有h时，模型在准确的低层特征形成前，就只关注比例c的训练样本，从而倾向于对错误的子集过拟合。辅助头让主干在整个训练过程中接触全部样本。例如，α取得很小时，SelectiveNet会产生较差的选择性风险。所有实验均采用$\alpha = 0.5$，没有做超参数优化。

5 覆盖率准确性

对于式（1）这样的带约束经验风险最小化（ERM），预期测试集上会出现约束违反。在本文中，即真实覆盖率$\phi(g)$可能低于目标c。另一方面，即使不违反约束，也可能得到$\phi(g) > c$的次优模型(f, g)（最优选择性模型必须以等号满足约束）。深度模型的高容量可能放大覆盖率不准确的问题。

译注：“必须取等号”是原文概括，通常需要可调整选择规则等条件。在风险平坦、原子分布或受限假设类下，不能无条件断言所有最优模型均取等号。

表1给出SelectiveNet和SR的测试覆盖率。SelectiveNet以目标$c \in \{0.7, 0.75, \ldots, 0.9\}$优化，SR则在训练集上选适当拒绝阈值来控制覆盖率。两者在所有情形均偏离目标。例如目标为75%时，SR为64.16%，SelectiveNet为80.17%。

译注：原文列出的集合止于0.9，但表1还包含0.95。此处均保留。原文的“违反”也包含高于目标的偏离，不全是违反单边约束$\phi \geq c$。

表 1: CIFAR-10上不同目标下的测试覆盖率（%）。

目标覆盖率	SR测试覆盖率	SelectiveNet测试覆盖率
0.70	60.19 ± 0.23	77.34 ± 0.05
0.75	64.16 ± 0.26	80.17 ± 0.11
0.80	68.01 ± 0.36	82.63 ± 0.12
0.85	72.05 ± 0.30	85.34 ± 0.24
0.90	76.41 ± 0.26	87.74 ± 0.32
0.95	82.30 ± 0.23	90.99 ± 0.67
平均偏离	11.98	3.63

虽然SelectiveNet比SR的覆盖率准确性显著更好（平均偏离3.625%对11.98%，见表1），我们仍希望测试覆盖率尽可能接近目标。一个简单的训练后覆盖率校准方法即可做到，它使用独立的无标签验证集V_n，包含n个样本。估计$g(x)$的适当阈值τ，按如下规则预测：

$$(f, g_\tau)(x) \triangleq \begin{cases} f(x), & g(x) \geq \tau; \\ \text{不知道}, & \text{其他情形}. \end{cases}$$

给定V_n，把τ设为$g(x_i)$（$x_i \in V_n$）分布的第$100(1-c)$百分位数。注意事件$g(x_i) \geq \tau$为伯努利变量，直接应用Hoeffding界（1963），可对双侧覆盖率偏离超过ϵ的概率给出界。原文为

$$\Pr\{\epsilon\text{-偏离}\} \leq 2e^{2n\epsilon^2}.$$

令右侧等于置信参数δ，原文得到：以至少$1-\delta$的概率，模型覆盖率位于$[c-\epsilon, c+\epsilon]$，其中

$$\epsilon = \sqrt{\frac{\ln(2/\delta)}{2n}}.$$

译注：原文指数漏了负号；常见Hoeffding形式为$2e^{-2n\epsilon^2}$，才可推出后式。另因阈值由同一校准集选取，直接套用固定阈值的Hoeffding界并不充分；严格论证应使用经验分布一致界（如D-

KW）或独立评估集，并处理有限样本分位数误差与并列分数。上述是原文推导，非本译本新增保证。

6 实验设计与细节

完整代码见https://github.com/geifmany/SelectiveNet。本节描述数据集、对比基线、架构与超参数。

6.1 数据集

街景门牌号（SVHN） SVHN（Netzer等，2011）包含73,257幅训练图像、26,032幅测试图像，分为10个数字类别。来自Google街景的门牌数字经裁剪、对齐，图像尺寸为$32 \times 32 \times 3$像素。

CIFAR-10 CIFAR-10（Krizhevsky与Hinton，2009）含50,000幅训练图像和10,000幅测试图像，分为10类，图像尺寸为$32 \times 32 \times 3$像素（RGB）。

猫狗（Cats vs. Dogs） 取自ASIRRA数据集，含25,000幅猫狗图像，每类12,500幅。我们按分层方式随机分成20,000幅训练图像和5,000幅测试图像，均缩放至64×64。原始图像平均尺寸约为360×400。

混凝土抗压强度 来自UCI仓库（Dheeru与Karra Taniskidou，2017）的回归数据集，含1,030个样本、8个数值特征和1个目标值。目标为混凝土抗压强度，预测依据为成分（7个特征）与龄期（1个特征）。

6.2 基线

比较SR与MC-dropout两种基线。

Softmax响应（SR） 对给定样本，用softmax最大激活值估计预测置信度。Geifman与El-Yaniv（2017）指出，这一最直接的方法也是选择性预测中表现最佳者之一。

蒙特卡洛dropout Gal与Ghahramani（2016）的方法通过样本在启用dropout的网络中多次前向传播的方差统计量量化置信度。分类与回归均按其作者建议实现。分类使用dropout率$p = 0.5$及100次前向蒙特卡洛迭代；回归使用$p = 0.05$及200次。

6.3 架构与超参数

CNN实验使用著名的VGG-16（Simonyan与Zisserman，2014），按Liu与Deng（2015）的建议适配小数据集与小图像。相比原版：（i）只用一个512神经元全连接层，而非两个4096神经元层；（ii）加入批归一化（Ioffe与Szegedy，2015）；（iii）加入dropout（Srivastava等，2014）。标准增强包含水平翻转、垂直与水平平移和旋转。网络用SGD优化，动量0.9、初始学习率0.1、权重衰减5×10^{-4}；每25个epoch学习率减半，共300个epoch。该设置下全覆盖验证准确率为SVHN 96.79%、CIFAR-10 93.21%、猫狗96.42%，接近VGG-16已知最佳结果。

SelectiveNet主干为上述VGG-16（见图1）。预测头f与辅助头h均为全连接softmax层。选择头g含一个512神经元全连接隐层、批归一化和ReLU，再连接单个sigmoid输出神经元。所有实验$\alpha = 0.5$、$\lambda = 32$；该λ足够大，能在训练过程中维持约束。

UCI表格回归实验采用全连接主干，含一个64神经元ReLU隐层。预测与辅助头均为单个线性激活神经元的全连接层。选择头g含16神经元ReLU隐层，随后是单个sigmoid神经元全连接层。所有模块（主干、预测、辅助、选择头）的每个隐层均用批归一化。采用ADAM（Kingma与Ba，2014），学习率5×10^{-4}，小批量256，训练800个epoch。使用平方损失，权重衰减10^{-4}。

7 实验

本节报告验证SelectiveNet有效性的实验。先做三个主干为CNN的图像分类实验，再做主干为全连接网络的回归实验。

7.1 选择性分类

首先考虑CIFAR-10。我们训练主干为VGG-16的SelectiveNet，完整架构和实现见第6节。比较按$c \in \{0.7, 0.75, 0.8, 0.85, 0.9, 0.95\}$训练并校准后的SelectiveNet与两个标准选择性分类器。后者在同一主干上添加与f相同的多类别头，使用SR与MC-dropout置信度估计，按照Geifman与El-Yaniv（2017）在CIFAR-10全覆盖条件下训练。

表2显示SelectiveNet显著优于两个基线。相对优势从覆盖率0.95时比SR好8.5%，到覆盖率0.75时比MC-dropout好26.8%。MC-dropout与SR数值相近，与Geifman与El-Yaniv（2017）的报告一致。 接着是SVHN。使用相同VGG-16架构，目标$c \in \{0.8, 0.85, 0.9, 0.95\}$；覆盖率低于

表 2: CIFAR-10分类：各覆盖率下选择性风险（0/1错误率，%）。改善为相对于对应基线的百分比。

覆盖率	SelectiveNet	MC-dropout	改善%	SR	改善%
1.00	6.79 ± 0.03	6.79 ± 0.03	0.00	6.79 ± 0.03	0.00
0.95	4.16 ± 0.09	4.58 ± 0.05	8.98	4.55 ± 0.07	8.56
0.90	2.43 ± 0.08	2.92 ± 0.01	16.99	2.89 ± 0.03	16.14
0.85	1.43 ± 0.08	1.82 ± 0.09	21.35	1.78 ± 0.09	19.79
0.80	0.86 ± 0.06	1.08 ± 0.05	20.39	1.05 ± 0.07	17.87
0.75	0.48 ± 0.02	0.66 ± 0.05	26.86	0.63 ± 0.04	22.71
0.70	0.32 ± 0.01	0.43 ± 0.05	26.38	0.42 ± 0.06	23.88

0.8时，该数据集的模型已近乎完美。表3显示，除0.95时各方法统计上难区分外，SelectiveNet在所有覆盖率均显著领先。在0.8覆盖率下最大相对优势为14.07%。SR与MC-dropout在各覆盖率的表现仍很相似。

译注：正文所说最大14.07%与表3不一致；表3在0.85覆盖率下对MC-dropout的改善为15.26%。均保留原文。改善值也不按已四舍五入的展示均值重新计算。

最后，在图像较大（64×64）的猫狗数据集上，将同一架构为所有方法适配该尺寸，按SVHN相同覆盖率重复实验。表4再次显示SelectiveNet相对两个基线有一致、显著的优势。图2给出CIFAR-10和猫狗数据集上SelectiveNet与SR的风险—覆盖率曲线，清楚展示这种一致优势。

表 3: SVHN分类：各覆盖率下选择性风险（0/1错误率，%）。

覆盖率	SelectiveNet	MC-dropout	改善%	SR	改善%
1.00	3.21 ± 0.08	3.21 ± 0.08	0.00	3.21 ± 0.08	0.00
0.95	1.40 ± 0.01	1.40 ± 0.05	0.00	1.39 ± 0.05	-0.77
0.90	0.82 ± 0.01	0.90 ± 0.04	9.05	0.89 ± 0.04	8.47
0.85	0.60 ± 0.01	0.71 ± 0.03	15.26	0.70 ± 0.03	13.61
0.80	0.53 ± 0.01	0.61 ± 0.01	14.07	0.61 ± 0.02	14.07

表 4: 猫狗分类：各覆盖率下选择性风险（0/1错误率，%）。

覆盖率	SelectiveNet	MC-dropout	改善%	SR	改善%
1.00	3.58 ± 0.04	3.58 ± 0.04	0.00	3.58 ± 0.04	0.00
0.95	1.62 ± 0.05	1.92 ± 0.06	15.40	1.91 ± 0.08	15.09
0.90	0.93 ± 0.01	1.10 ± 0.05	16.13	1.10 ± 0.08	15.56
0.85	0.56 ± 0.02	0.78 ± 0.06	28.30	0.82 ± 0.06	32.40
0.80	0.35 ± 0.09	0.55 ± 0.02	36.39	0.68 ± 0.05	48.16

7.2 选择性回归

我们在UCI混凝土抗压强度数据集上训练适于表格回归的SelectiveNet变体。主干为全连接网络，预测头采用第6节的线性单元，结果见表5。回归网络没有softmax层，因此SR不能适用。我们只与MC-dropout比较，它是目前深度选择性回归唯一的非集成／贝叶斯方法。对所有低于0.9的覆盖率，SelectiveNet有明显优势，显著优于MC-dropout，相对优势为26.81%至30.48%。

译注： 26.81、30.48实际上是表5中的风险值；表中低于0.9覆盖率的改善为7.23%至17.09%。原文正文与表格不一致，译文分别保留。“非集成／贝叶斯”也保留原措辞，不作为当前技术分类依据。

表 5: 混凝土抗压强度回归：各覆盖率下选择性风险（MSE）。

覆盖率	SelectiveNet	MC-dropout	改善%
1.00	38.45 ± 0.90	39.01 ± 0.70	1.43
0.90	35.35 ± 1.31	36.97 ± 0.46	4.39
0.80	30.48 ± 0.93	35.53 ± 0.61	14.23
0.70	27.94 ± 1.12	33.70 ± 0.58	17.09
0.60	27.12 ± 0.99	31.23 ± 0.71	13.18
0.50	26.81 ± 1.36	28.90 ± 0.77	7.23

8 经验观察

现在从经验上分析SelectiveNet与校准和表示结构有关的性质。

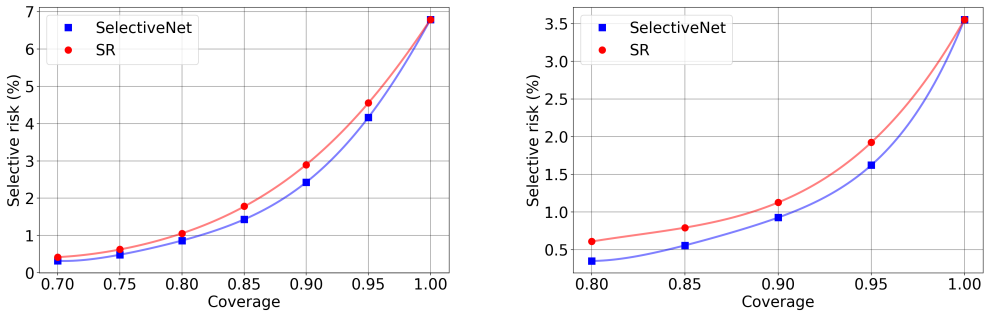
8.1 覆盖率校准

我们评估训练后校准的效果和局限。对CIFAR-10上按$c \in \{0.7, 0.75, 0.8, 0.85, 0.9\}$训练的五个模型，分别校准至每个覆盖率水平。表6是覆盖率“混淆矩阵”，元素(i, j)表示模型i校准到覆盖率j后的风险。例如，末列第三个值2.40表示按0.8训练、再校准至0.9的SelectiveNet的选择性风险。表中数字均乘以100，反映错误率百分比；每个风险值为三次独立运行的平均。 每列最小元

表 6: CIFAR-10选择性风险：行为训练覆盖率，列为校准覆盖率。数值乘100，单位为错误率%；红色粗体保留原文近最优标记。

训练／校准	0.70	0.75	0.80	0.85	0.90
0.70	**0.32 ± 0.01**	0.55 ± 0.00	0.88 ± 0.05	1.48 ± 0.05	2.65 ± 0.11
0.75	**0.31 ± 0.04**	**0.48 ± 0.02**	**0.85 ± 0.04**	1.47 ± 0.07	2.63 ± 0.06
0.80	**0.34 ± 0.01**	0.55 ± 0.04	**0.86 ± 0.06**	**1.42 ± 0.05**	**2.40 ± 0.09**
0.85	0.37 ± 0.02	0.54 ± 0.03	**0.84 ± 0.04**	**1.43 ± 0.08**	**2.47 ± 0.07**
0.90	0.51 ± 0.12	0.58 ± 0.09	**0.79 ± 0.03**	**1.35 ± 0.03**	**2.43 ± 0.08**

素用红色粗体显示；同列中与其在一个标准误范围内重叠的其他元素也如此标记。因此，红色元素代表该列全部近最优错误率。显然，对角线为红色粗体，表明按c训练的校准后SelectiveNet，在按其他覆盖率训练的模型中接近最优。另一个有趣观察是，只要向接近训练目标的覆盖率校准，就不会有显著损失。非对角元素常呈现更差风险，说明校准后的SelectiveNet确实优化了特定覆盖率的风险。



(a) Cifar-10 (b) Cats vs. dogs

图 2: SelectiveNet与SR的风险—覆盖率曲线：（a）CIFAR-10；（b）猫狗。Coverage为覆盖率，Selective risk为选择性风险，曲线保留原版。

8.2 学到的表示

观察SelectiveNet与SR、MC-dropout等标准方法学到的表示很有意思。由架构可预期，SelectiveNet主干会产生对预测和拒绝都有效的特征。标准方法则依赖不考虑覆盖率的普通模型。为此，使用t分布随机邻域嵌入（t-SNE；Maaten与Hinton，2008），可视化在CIFAR-10上以$c = 0.7$训练的SelectiveNet与SR的表示层。我们对嵌入层全部512个神经元构成的表示计算t-SNE变换，见图3。

图3（b）为SR：每种颜色对应10个CIFAR-10标签之一，如青绿色簇是狗，橙色簇是猫。拒绝样本为浓色点，被覆盖样本为淡色点。图3（a）同样展示SelectiveNet的t-SNE。

显著现象是，SR的簇不论是否拒绝均相对分开，而SelectiveNet的拒绝样本分离较弱。例如，飞机（栗色）、船（浅棕色）与鸟（蓝色）的拒绝点严重混淆；猫、狗、鹿、马、蛙等其余动物的拒绝点也没有良好分开。合理解释是，SelectiveNet未浪费表示容量去分离拒绝点，因而留下更多容量学习相关特征，区分非拒绝点。另外，多数拒绝点位于一个中心簇，这种结构可能让选择函数g更容易捕获这些点。



(a) SelectiveNet (b) Softmax response (c) Legend

图 3: （a）SelectiveNet与（b）SR的嵌入表示t-SNE可视化；（c）图例。浓色为拒绝，淡色为覆盖。图例类别依次为飞机、汽车、鸟、猫、鹿、狗、蛙、马、船、卡车；保留原图英文标签。

9 结语

我们提出SelectiveNet，一种有效的深度选择性分类和回归模型。其能力来自在同一网络中联合训练选择与预测模型，使其关注生产环境中不会拒绝的更相关样本。MC-dropout或SR等已知技术通常在全覆盖训练的普通模型上叠加拒绝机制，无法受益于这一优势。经验结果表明，在给定弃权率下，新模型目前提供最准确的分类。对于选择性回归，SelectiveNet提供最佳且唯一快速的解决方案，唯一其他但较差的替代是MC-dropout。这鼓励在视觉跟踪、检测等需要快速推理的深度回归应用中使用SelectiveNet。

未来研究仍有若干问题。与许多深度学习应用一样，集成可改善选择性预测的基础模型。Lakshminarayanan等（2017）最近表明，2至15个模型的集成可改善基本SR拒绝方法。检验集成能否同样改善SelectiveNet会很有意义。另一个方向是考察选择函数g的容量（架构选择）对整体表现的影响。本文未优化该选择，但预期不同数据集、乃至不同覆盖率，可能也需要不同选择头架构。

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参考文献与英文原文

以下附PMLR正式版完整9页；其最后一页为完整参考文献。为保证作者姓名、英文题名、出版物与页码可准确检索，书目保持原版。中文与原文若有歧义，应对照原文及本译本明确标记的译注。

SelectiveNet: A Deep Neural Network with an Integrated Reject Option

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Abstract

We consider the problem of *selective prediction* (also known as *reject option*) in deep neural networks, and introduce SelectiveNet, a deep neural architecture with an integrated reject option. Existing rejection mechanisms are based mostly on a threshold over the prediction confidence of a pre-trained network. In contrast, SelectiveNet is trained to optimize both classification (or regression) and rejection simultaneously, end-to-end. The result is a deep neural network that is optimized over the covered domain. In our experiments, we show a consistently improved risk–coverage trade-off over several well-known classification and regression datasets, thus reaching new state-of-the-art results for deep selective classification.

1. Introduction

Detecting and controlling statistical uncertainties in machine learning processes is essential in many mission-critical machine learning applications such as autonomous driving, medical diagnosis or home robotics. In *selective prediction* our goal is to learn predictive models that know what they do not know. These models are allowed to abstain whenever they are not sufficiently confident in their prediction.

The idea of a reject option was already studied over 60 years ago by Chow (1957). Reject option mechanisms have been considered for many hypothesis classes and learning algorithms. These mechanisms are mainly driven by two related ideas. The first is the cost-based model whereby the selective model is optimized to minimize a loss function that tracks a user-specified cost for abstention. Other mechanisms are built upon functions that serve as proxies for prediction confidence. Typical examples of these proxies are functional margin, distance to nearest neighbor(s), etc.

In this work we focus on selective models for deep neural networks (DNNs). The most recent relevant paper on this topic is Geifman & El-Yaniv (2017), which shows how to construct a probabilistically-calibrated selective classifier using any certainty estimation function for a given (already trained) model. Interestingly, among the best certainty proxies for a trained network is the simple Softmax Response (SR) estimator, which is often used by practitioners.

Within an abstention-constrained framework, whereby the user specifies the desired target coverage, we optimize, end-to-end, a neural network with an integrated reject option, designed to be optimal for the required coverage slice. Our technique yields effective selective models that consistently improve all known techniques.

From a learning-theoretic perspective, it is not hard to argue that learning a specialized rejection model for each coverage slice is potentially better than rejecting based on confidence rates extracted from a pre-trained network. To demonstrate the advantage of such specialized training, consider the following vignette. Bob and Alice are about to take an exam at the end of their machine learning course. It is announced that the exam will contains 10 questions, one for each of the course topics, but the students will be asked to choose exactly five questions. Alice decides to cover only the easiest five topics while Bob's mom forces him to study all ten topics. Who do you think is going to get the higher grade?

The contributions of this paper are:

- A selective loss function that optimizes a specified coverage slice using a variant of the interior point optimization method (Potra & Wright, 2000).
- SelectiveNet: A three-headed network for end-to-end learning of selective classification models.
- SelectiveNet for regression: Providing the first alternative to costly methods such as MC-dropout or ensemble techniques.
- An empirical advantage demonstrated over four datasets for classification and regression with a significant advantage over MC-dropout and SR.

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2. Selective Prediction Problem Formulation

In this work we consider selective prediction problems of any type, classification or regression. A supervised prediction task is formulated as follows. Let $\mathcal{X}$ be any feature space and $\mathcal{Y}$ a label space. For example, in classification, $\mathcal{X}$ could be images, and $\mathcal{Y}$ could be class labels in the case of classification or the coordinates of a bounding box in the case of regression. Let $P(X,Y)$ be a distribution over $\mathcal{X}\times\mathcal{Y}$. A model, $f:\mathcal{X}\to\mathcal{Y}$, is called a *prediction function*, and its *true risk* w.r.t. P is $R(f)\triangleq E_{P(X,Y)}[\ell(f(x),y)]$, where $\ell: Y\times Y\to\mathbb{R}^+$ is a given loss function, for example, the 0/1 error or the squared loss. Given a labeled set $S_m=\{(x_i,y_i)\}_{i=1}^m\subseteq(\mathcal{X}\times\mathcal{Y})^m$ sampled i.i.d. from $P(X,Y)$, the *empirical risk* of the classifier f is $\hat{r}(f|S_m)\triangleq\frac{1}{m}\sum_{i=1}^m\ell(f(x_i),y_i)$.

A *selective model* (El-Yaniv & Wiener, 2010) is a pair (f,g), where f is a *prediction function*, and $g:\mathcal{X}\to\{0,1\}$ is a *selection function*, which serves as a binary qualifier for f as follows,

$$(f,g)(x)\triangleq\begin{cases}f(x), & \text{if } g(x)=1;\\ \text{don't know}, & \text{if } g(x)=0.\end{cases}$$

Thus, the selective model abstains from prediction at a point x iff $g(x)=0$. A *soft* selection function can also be considered, where $g:\mathcal{X}\to[0,1]$, and decisions can be taken probabilistically or deterministically (w.r.t. a threshold). The performance of a selective model is quantified using *coverage* and *risk*. Coverage is defined to be

$$\phi(g)\triangleq E_P[g(x)],$$

the probability mass of the non-rejected region in $\mathcal{X}$. The selective risk of (f,g) is

$$R(f,g)\triangleq\frac{E_P[\ell(f(x),y)g(x)]}{\phi(g)}.$$

Clearly, the risk of a selective model can be traded-off for coverage. The entire performance profile of such a model can be specified by its *risk–coverage* curve, defined to be the risk as a function of coverage (El-Yaniv & Wiener, 2010; Wiener & El-Yaniv, 2012).

The true selective risk and true coverage have corresponding empirical counterparts that can be calculated for any given labeled set S_m. The *empirical selective risk* is

$$\hat{r}(f,g|S_m)\triangleq\frac{\frac{1}{m}\sum_{i=1}^m\ell(f(x_i),y_i)g(x_i)}{\hat{\phi}(g|S_m)},$$

and the *empirical coverage* is

$$\hat{\phi}(g|S_m)\triangleq\frac{1}{m}\sum_{i=1}^m g(x_i).$$

An optimal selective model could be defined in one of two ways: by either optimizing the selective risk, given a constraint on the coverage or vice versa. In this work, we focus on the first case, which is defined as follows. For a given coverage rate $0<c\leq 1$ and hypothesis class Θ, the optimal selective model is

$$\theta^*=\arg\min_{\theta\in\Theta}(R(f_\theta,g_\theta)) \quad s.t.\ \phi(g_\theta)\geq c. \tag{1}$$

In this paper, Θ is a set of parameters for a given deep network architecture for f and g.

It is possible to convert any selective model defined as above (with controlled coverage and optimized risk) to a model that controls risk and optimizes coverage using the technique of Geifman & El-Yaniv (2017).

3. Related Work

The literature on reject option techniques is quite extensive and mostly focuses on hypothesis classes and learning algorithms such as SVMs, nearest neighbours and boosting (Fumera & Roli, 2002; Hellman, 1970; Cortes et al., 2016; El-Yaniv & Wiener, 2015). When considering selective prediction in neural networks (NNs), a straightforward and effective technique is to set a threshold over a confidence score derived from a pre-trained network, which was already optimized to predict all points (Cordella et al., 1995; De Stefano et al., 2000; Wiener & El-Yaniv, 2012). Geifman & El-Yaniv (2017), extending this technique for deep neural networks (DNNs), show how to derive a selective classifier based on any confidence score function. They also show how to compute a high confidence guarantee on the selective risk at test time. In the latter work, two techniques for extracting confidence scores from an NN are discussed. The first is the Softmax Response (SR), defined to be the maximal activation in the softmax layer (in classification architectures). The second confidence score is the Monte-Carlo dropout (MC-dropout) technique of Gal & Ghahramani (2016). MC-dropout estimates prediction confidence based on the statistics of numerous feed-forward passes through the network with dropout applied. To the best of our knowledge, the MC-dropout is the only non-Bayesian technique to extract a confidence score for a single network. Unfortunately, the prediction cost of an effective MC-dropout can reach hundreds of feed-forward iterations for each prediction.

Another family of confidence score functions that can be used for selective classification is based on statistics of ensembles of multiple models (Lakshminarayanan et al., 2017; Geifman et al., 2018). In the present work, we focus on selective classification with a single classifier while noting that extending our work for ensembles of selective classifiers is

straightforward and likely to improve the results.

4. SelectiveNet

In this section we first define *SelectiveNet*, a deep neural architecture allowing end-to-end optimization of selective models. In conjunction, we also propose a suitable loss function and optimization procedure to train it for any desired target coverage.

4.1. SelectiveNet Architecture

SelectiveNet is a selective model (f, g) that optimizes both $f(x)$ and $g(x)$ in a single DNN model. A schematic view of SelectiveNet is depicted in Figure 1. The input is processed by the main body block, consisting of a number of DNN layers or sub-blocks. The last layer of the main body is termed the *representation layer*. The main body block can be assembled using any type of architecture relevant to the problem at hand (e.g., convolutional, fully connected or recurrent architectures). SelectiveNet has three output heads for prediction, selection and auxiliary prediction. The role of the selection head is to implement the selection function $g(x)$. The role of the prediction head is to implement the prediction $f(x)$, and the auxiliary prediction head (denoted as $h(x)$) predicts a related prediction task in order to enrich or enforce the construction of relevant features in the main body. The auxiliary head is only used for training and its role will be clarified later after we define the loss function. The architectures of these three heads can vary depending on the task type and complexity but it is always the case that the final layer of the selection head $g(x)$ is a single neuron with a sigmoid activation. The final layer of $f(x)$ depends on the application and could be softmax (classification) or linear (regression). At inference time, a sample x is fed to SelectiveNet, which predicts $f(x)$ if and only if $g(x) \geq 0.5$; otherwise, SelectiveNet abstains from predicting the label of x.

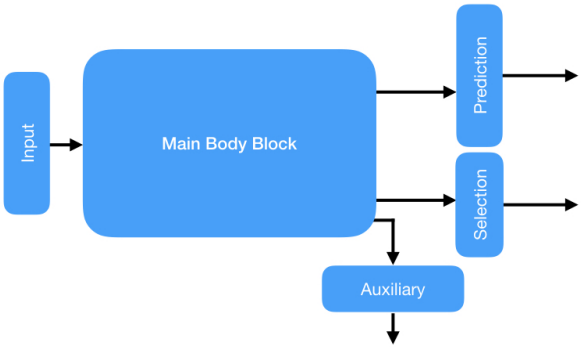


Figure 1. The SelectiveNet schematic architecture

4.2. SelectiveNet Optimization

The selective prediction objective is given in Equation (1). To enforce the coverage constraint, we utilize a variant of the well-known Interior Point Method (IPM) (Potra & Wright, 2000). This results in the following unconstrained objective, which is averaged over the samples in S_m,

$$\begin{aligned} \mathcal{L}_{(f,g)} &\triangleq \hat{r}_\ell(f, g|S_m) + \lambda\Psi(c - \hat{\phi}(g|S_m)) \qquad (2) \\ \Psi(a) &\triangleq \max(0, a)^2, \end{aligned}$$

where c is the target coverage, λ is a hyperparameter controlling the relative importance of the constraint, and Ψ is a quadratic penalty function. The choice of λ is discussed in Section 6. In this work we train the auxiliary head, h, using the same prediction task as assigned to f (classification or regression) using a standard loss function,

$$\mathcal{L}_h = \hat{r}(h|S_m) = \frac{1}{m}\sum_{i=1}^{m} \ell(h(x_i), y_i),$$

which is oblivious to any coverage considerations. We optimize SelectiveNet using a convex combination of the selective loss $\mathcal{L}_{(f,g)}$ (2) and the auxiliary loss $\mathcal{L}_h$. Thus, the overall training objective is

$$\mathcal{L} = \alpha\mathcal{L}_{(f,g)} + (1 - \alpha)\mathcal{L}_h.$$

The use of the auxiliary head, h, is essential to optimizing SelectiveNet. Without h, SelectiveNet will focus on a fraction c of the training set, before accurate low level features are constructed. In such a case, SelectiveNet will tend to overfit to the wrong subset of the training set. The auxiliary head exposes the main body block to all training instances throughout the training process. For example, when taking the parameter alpha to be very small, SelectiveNet will result with poor selective risk. In all our experiment we used $\alpha = 0.5$ without any hyperparameter optimization.

5. Coverage Accuracy

When considering learning with constrained ERM as in Equation (1), we expect to see violations of the constraint over a test set. In our case, this means that the true coverage, $\phi(g)$, might be smaller than the desired target coverage c. On the other hand, even if the constraint is not violated, we are susceptible to sub-optimal models, (f, g), where $\phi(g) > c$ (the optimal selective model must satisfy the constraint with equality). When dealing with deep neural models, such coverage inaccuracies can be amplified due to the high capacity of these models.

Indeed, consider Table 1 showing the resulting test coverage obtained using both SelectiveNet and SR. Here, SelectiveNet was optimized with target coverage rates $c \in$

Target coverage	SR test coverage	SelectiveNet test coverage
0.70	60.19±0.23	77.34±0.05
0.75	64.16±0.26	80.17±0.11
0.80	68.01±0.36	82.63±0.12
0.85	72.05±0.30	85.34±0.24
0.90	76.41±0.26	87.74±0.32
0.95	82.30±0.23	90.99±0.67
Average violation	11.98	3.63

Table 1. Selective test coverage rates (%) for various trained coverage rates for SR and SelectiveNet for the Cifar-10 dataset.

$\{0.7, 0.75, \ldots, 0.9\}$, and the coverage of SR was controlled by choosing an appropriate rejection threshold over the training set. Clearly, both SR and SelectiveNet violate the target coverage rate in all cases. For example, when the target coverage is 75%, the realized coverage of SR is 64.16% and that of SelectiveNet is 80.17%.

While the coverage accuracy of SelectiveNet is significantly better than that of SR (average violation 3.625% vs. 11.98%, see Table 1), we are interested in generating selective models whose test coverage is as close as possible to the target coverage. This goal can be achieved easily using the following simple post-training coverage calibration technique, which relies on an independent *unlabeled* validation set V_n containing n samples. To calibrate to the target coverage, we estimate an appropriate threshold τ for $g(x)$ values and predict using the following decision rule

$$(f, g_\tau)(x) \triangleq \begin{cases} f(x), & \text{if } g(x) \geq \tau \\ \text{don't know}, & \text{otherwise.} \end{cases}$$

Given the validation set V_n, τ is set to be the $100(1-c)$ percentile of the distribution of $g(x_i)$, $x_i \in V_n$.

Noting that the event $g(x_i) \geq \tau$ is a Bernoulli variable, we can straightforwardly use the Hoeffding bound (Hoeffding, 1963) to bound the probability of (double sided) coverage violation greater than ϵ,

$$\mathbf{Pr}\{\epsilon\text{−violation}\} \leq 2e^{2n\epsilon^2}.$$

Equating the right-hand side to a confidence parameter δ, we get that with probability at least $1-\delta$, the resulting selective model will be in the range $[c-\epsilon, c+\epsilon]$ where $\epsilon = \sqrt{\ln(2/\delta)/(2n)}$.

6. Experimental Design and Details

Our complete code can be downloaded from the following link, https://github.com/geifmany/SelectiveNet. This section describes the experimental details: datasets used, baseline algorithms used for comparison purposes, and our choice of architectures and hyperparamers.

6.1. Datasets

Street View House Numbers (SVHN). The SVHN dataset (Netzer et al., 2011) is an image classification dataset containing 73,257 training images and 26,032 test images classified into 10 classes representing digits. The images are digits of house street numbers, which are cropped and aligned, taken from the Google Street View service. Image size is $32 \times 32 \times 3$ pixels.

CIFAR-10. The CIFAR-10 dataset (Krizhevsky & Hinton, 2009) is an image classification dataset comprising a training set of 50,000 images and 10,000 test images classified into 10 categories. The image size is $32 \times 32 \times 3$ pixels (RGB images).

Cats vs. Dogs. The Cats vs. Dogs is an image classification dataset extracted from the ASIRRA dataset. It contains 25,000 images of cats and dogs, 12,500 in each class. We randomly split this dataset into a training set containing 20,000 images, and a test set of 5000 images in a stratified fashion. We also rescaled each image to size 64x64. The average dimension size of the original images is roughly 360x400.

Concrete Compressive Strength. The Concrete Compressive Strength dataset is a regression dataset from the UCI repository (Dheeru & Karra Taniskidou, 2017). It contains 1030 instances with eight numerical features and one target value. The target value is the compressive strength of concrete, which is predicted based on its ingredients (seven features) and age (one feature).

6.2. Baseline Methods

We compare the proposed method with two baselines: SR and MC-dropout, which are described below.

Softmax Response (SR). The SR method estimates prediction confidence by the maximum softmax value activation for a given instance. According to Geifman & El-Yaniv (2017), SR, which is the most straightforward method, is a top performer in selective prediction.

Coverage	SelectiveNet risk	MC-dropout risk	% improvement	SR risk	% improvement
1.00	6.79±0.03	6.79±0.03	0.00	6.79±0.03	0.00
0.95	**4.16±0.09**	4.58±0.05	8.98	4.55±0.07	8.56
0.90	**2.43±0.08**	2.92±0.01	16.99	2.89±0.03	16.14
0.85	**1.43±0.08**	1.82±0.09	21.35	1.78±0.09	19.79
0.80	**0.86±0.06**	1.08±0.05	20.39	1.05±0.07	17.87
0.75	**0.48±0.02**	0.66±0.05	26.86	0.63±0.04	22.71
0.70	**0.32±0.01**	0.43±0.05	26.38	0.42±0.06	23.88

Table 2. Classification experiment with **Cifar-10**. Selective risk (0-1% error) for various coverage rates.

Monte Carlo-dropout (MC-Dropout) (Gal & Ghahramani, 2016). Prediction confidence is quantified using variance statistics of multiple feed-forward passes of an instance through the network with dropout applied. In both classification and regression, MC-dropout was implemented as recommended by its creators. Specifically, in classification, we used a dropout rate of $p = 0.5$ and 100 feed-forward MC iterations. In regression, we used a dropout rate of $p = 0.05$ and 200 feed-forward MC iterations.

6.3. Architectures and Hyperparameters

For the convolutional neural network (CNN) experiments, we used the well-known VGG-16 architecture (Simonyan & Zisserman, 2014), optimized for the small datasets and image sizes as suggested by Liu & Deng (2015). The changes from the original VGG-16 architecture are: (i) we used only one fully connected layer with 512 neurons (the original VGG-16 has two fully connected layers of 4096 neurons). (ii) we added batch normalization (Ioffe & Szegedy, 2015) (iii) we added dropout (Srivastava et al., 2014). We used standard data augmentation consisting of horizontal flips, vertical and horizontal shifts, and rotations. The network was optimized using *stochastic gradient descent* (SGD) with a momentum of 0.9, an initial learning rate of 0.1, and a weight decay of 5e-4. The learning rate was reduced by 0.5 every 25 epochs, and trained for 300 epochs. With this setting, we reached *full coverage* validation accuracy of 96.79% for SVHN, 93.21% for Cifar-10 and 96.42% for Cats vs. Dogs. These are close to the best known results for the VGG-16 architecture.

The main body block of SelectiveNet is the VGG-16 architecture (see Figure 1). Both the prediction (f) and auxiliary heads (h) of SelectiveNet are fully-connected softmax layers. The selection head (g) is a fully connected hidden layer with 512 neurons, batch normalization and ReLU activation, followed by a fully connected layer to one output neuron with sigmoid activation. The value of α (the convex combination between the selective loss and the auxiliary loss) was set to 0.5 for all experiments, and λ was set to 32 (this value was found to be large enough to preserve the constraint through the training process).

In the regression experiment, which considers a tabular dataset from the UCI repository, we used a fully connected network as the main body block. This block consists of one hidden layer of 64 ReLU activated neurons. Both the prediction and auxiliary heads are fully connected layers with one linearly activated neuron. The selection head g was computed with one hidden layer, with 16 ReLU activated neurons, followed by a fully connected layer with one Sigmoid activated neuron. Batch normalization was applied in every hidden layer on all modules (main body block, prediction, auxiliary and selection heads). The regression model was optimized using the ADAM algorithm (Kingma & Ba, 2014) with a learning rate of 5×10^{-4} and mini-batch size of 256 and 800 epochs. We used squared loss with weight decay of 1×10^{-4} during optimization.

7. Experiments

In this section we report on the results of several experiments that validate the effectiveness of SelectiveNet. We begin with three image classification experiments where the main body block (see Section 4) has a CNN architecture. We then experiment with a regression task where the main body block comprises a fully connected architecture.

7.1. Selective Classification

We consider the Cifar-10 dataset first. We trained a version of SelectiveNet whose main body block is based on the VGG-16 architecture; see the full description of our architecture and implementation details in Section 6. We compared calibrated SelectiveNet instances, which were trained with coverage rates $c \in \{0.7, 0.75, 0.8, 0.85, 0.9, 0.95\}$ to two standard selective classifiers over these coverage rates. The standard classifiers were constructed using the well-known SR and MC-dropout confidence estimates applied to a trained network consisting of a multi-class head (identical to f in SelectiveNet) on top of the same main body block of SelectiveNet. The standard model was trained on Cifar-10 at full coverage as proposed in (Geifman & El-Yaniv,

Coverage	SelectiveNet risk	MC-dropout risk	% improvement	SR risk	% improvement
1.00	3.21±0.08	3.21±0.08	0.00	3.21±0.08	0.00
0.95	1.40±0.01	1.40±0.05	0.00	1.39±0.05	-0.77
0.90	**0.82±0.01**	0.90±0.04	9.05	0.89±0.04	8.47
0.85	**0.60±0.01**	0.71±0.03	15.26	0.70±0.03	13.61
0.80	**0.53±0.01**	0.61±0.01	14.07	0.61±0.02	14.07

Table 3. Classification experiment with **SVHN**. Selective risk (0-1% error) for various coverage rates.

Coverage	SelectiveNet risk	MC-dropout risk	% improvement	SR risk	% improvement
1.00	3.58±0.04	3.58±0.04	0.00	3.58±0.04	0.00
0.95	**1.62±0.05**	1.92±0.06	15.40	1.91±0.08	15.09
0.90	**0.93±0.01**	1.10±0.05	16.13	1.10±0.08	15.56
0.85	**0.56±0.02**	0.78±0.06	28.30	0.82±0.06	32.40
0.80	**0.35±0.09**	0.55±0.02	36.39	0.68±0.05	48.16

Table 4. Classification experiment with **Cats vs. Dogs**. Selective risk (0-1 % error) for various coverage rates.

2017). The results are presented in Table 2. It can be seen that SelectiveNet significantly outperforms the two baseline methods (SR and MC-dropout). The relative advantage of SelectiveNet varies from 8.5% at 0.95 coverage compared to SR, and a maximal advantage of 26.8% compared to MC-dropout at 0.75 coverage. It is also evident that the MC-dropout and SR performance is quantitatively similar as was already reported in (Geifman & El-Yaniv, 2017).

We turn now to the SVHN dataset. Here again, we trained SelectiveNet with the same VGG-16-based architecture for coverage rates $c \in \{0.8, 0.85, 0.9, 0.95\}$ (coverage rates smaller than 0.8 result in nearly perfect models in this dataset). The results in Table 3 show SelectiveNet's significant superiority for all coverage rates, with the exception of the 0.95 coverage, where all methods are statistically indistinguishable. The maximal relative advantage over this dataset is 14.07% for 0.8 coverage rate. We also observe that here again, the performance of SR and MC-dropout is similar for all coverage rates.

Finally, for the Cats vs. Dogs dataset, whose input image size is larger (64×64), we adapted the same architecture (to this image size) for all the contenders and repeated the same experiments with the same coverage rates as in the SVHN experiment. Presented in Table 4, the results are unsurprising and again indicate a consistent and significant advantage of SelectiveNet over the two baselines.

Risk-coverage curves depicting the performance of SelectiveNet and SR for the Cifar-10 and Cats vs. Dogs datasets appear in Figure 2. These curves nicely demonstrate the consistent advantage of SelectiveNet.

7.2. Selective Regression

We ran the regression experiment over the Concrete Compressive Strength dataset from the UCI repository. For this set we trained a SelectiveNet variant adapted for a regression of tabular data. The main body block is now a fully connected neural network and the prediction heads utilize a linear unit as described in Section 6. The results appear in Table 5. Clearly, the SR baseline cannot be applied in this setting because there is no softmax layer in regression networks. We, therefore, compared SelectiveNet only to the MC-dropout method, currently the only non-ensemble/Bayesian method for deep selective regression. In this task, SelectiveNet exhibits a distinct advantage over MC-dropout for all coverage rates below 0.9. There, SelectiveNet significantly outperforms MC-dropout, with a relative advantage that varies between 26.81% to 30.48%.

8. Empirical Observations

We now empirically analyze some properties of SelectiveNet, related to calibration and representation structure.

8.1. Coverage Calibration

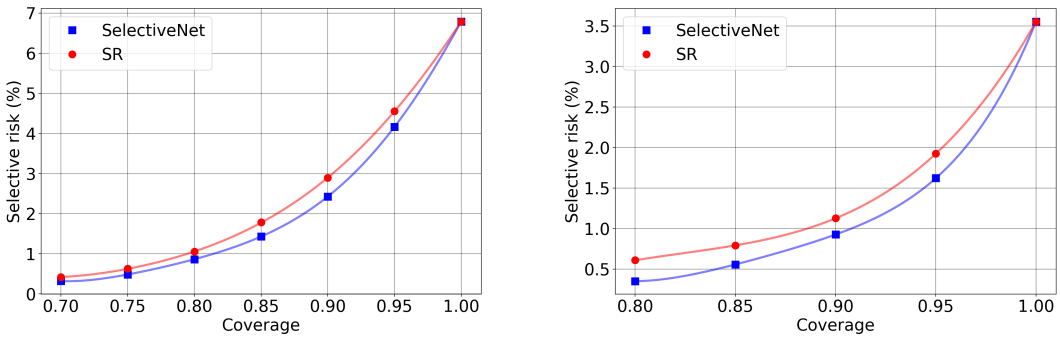
We evaluate the effectiveness and limitations of the post-training coverage calibration technique. We used this calibration to evaluate five models for Cifar-10 trained with five different coverage rates, $c \in \{0.7, 0.75, 0.8, 0.85, 0.9\}$. We calibrated each model to match each coverage value. Table 6 is the resulting coverage "confusion matrix" where element (i, j) in the table is the selective risk of model i calibrated to coverage j. For example, the third value in the last column (2.40) is the selective risk of SelectiveNet

Coverage	SelectiveNet risk	MC-dropout risk	% improvement
1.00	38.45±0.90	39.01±0.70	1.43
0.90	35.35±1.31	36.97±0.46	4.39
0.80	**30.48±0.93**	35.53±0.61	14.23
0.70	**27.94±1.12**	33.70±0.58	17.09
0.60	**27.12±0.99**	31.23±0.71	13.18
0.50	**26.81±1.36**	28.90±0.77	7.23

Table 5. Regression experiment with **Concrete Compressive Strength**. Selective risk (MSE) for various coverage rates.

	Calibration				
Training	0.70	0.75	0.80	0.85	0.90
0.70	$\mathbf{0.32 \pm 0.01}$	0.55 ± 0.00	0.88 ± 0.05	1.48 ± 0.05	2.65 ± 0.11
0.75	$\mathbf{0.31 \pm 0.04}$	$\mathbf{0.48 \pm 0.02}$	$\mathbf{0.85 \pm 0.04}$	1.47 ± 0.07	2.63 ± 0.06
0.80	$\mathbf{0.34 \pm 0.01}$	0.55 ± 0.04	$\mathbf{0.86 \pm 0.06}$	$\mathbf{1.42 \pm 0.05}$	$\mathbf{2.40 \pm 0.09}$
0.85	0.37 ± 0.02	0.54 ± 0.03	$\mathbf{0.84 \pm 0.04}$	$\mathbf{1.43 \pm 0.08}$	$\mathbf{2.47 \pm 0.07}$
0.90	0.51 ± 0.12	0.58 ± 0.09	$\mathbf{0.79 \pm 0.03}$	$\mathbf{1.35 \pm 0.03}$	$\mathbf{2.43 \pm 0.08}$

Table 6. Selective risk of SelectiveNet trained on the Cifar-10 dataset where rows represent a training coverage value and columns represent post-training calibration coverage values. All results were multiplied by 100 and correspond to % of errors.



(a) Cifar-10 (b) Cats vs. dogs

Figure 2. Risk coverage curves comparing SelectiveNet and SR over (a) Cifar-10 and (b) Cats vs. Dogs

that was optimized with coverage 0.8 during training and then calibrated for coverage 0.9. All the numbers inside the table were multiplied by 100 and reflect the error percent. Each selective risk in the table is an average over three independent runs.

The minimal element in each column appears in bold red, and all other elements in that column, which have one standard error overlap with it, appear in bold red as well. Thus, the red elements in each column represent all nearly-optimal errors in their column. Clearly, the diagonal of this matrix is bold red, indicating that the calibrated SelectiveNet, which was trained for coverage c, is nearly optimal among the other SelectiveNets that were trained with coverage rates other than c. Another interesting observation is that there is no significant compromise due to the calibration process, provided that we calibrate towards a coverage rate that is close to the trained (target) coverage. The fact that the off-diagonal elements often tend to admit inferior selective risks indicates that the calibrated SelectiveNet effectively optimizes its coverage-specific selective risk.

8.2. Learned Representation

It is interesting to inspect the representation learned by SelectiveNet and those learned by standard methods such as SR or MC-dropout. Considering SelectiveNet's architecture, one may expect that the main body block will produce fea-



(a) SelectiveNet (b) Softmax response (c) Legend

Figure 3. t-SNE visualization of the embedding representation of (a) SelectiveNet and (b) SR.

tures that are effective for both the prediction and rejection tasks. In contrast, the standard method relies on standard models that are oblivious to coverage considerations. To this end, we use the well-known t-Distributed Stochastic Neighbor Embedding (t-SNE) technique (Maaten & Hinton, 2008) to visualize the learned representation layer of SelectiveNet and SR that were trained for the Cifar-10 dataset with $c = 0.7$. We computed the t-SNE transformation of the embedding representation consisting of all 512 neurons in the embedding layer. The results are depicted in Figure 3. In Figure 3(b) we see the t-SNE of the SR model. Each color in the figure corresponds to a class label (among the 10 classes in Cifar-10). For example, the turquoise cluster is 'dog' and the orange cluster corresponds to 'cat'. Rejected instances appear as dots in bold colors and the covered points are faded dots. Similarly, Figure 3(a) presents the t-SNE of SelectiveNet.

The striking observation is that the SR clusters are relatively well separated regardless of rejection but, in contrast, the rejected instances in the SelectiveNet case are very weakly separated. For example, we see severe confusion between rejected points of classes 'airplane' (maroon), 'ship' (light brown), and 'bird' (blue). Similarly, we see that the rejected instances of the rest of the animals ('cat', 'dog', 'deer', 'horse' and 'frog') are not well separated as well. A plausible interpretation of this phenomenon is that SelectiveNet is not wasting representational capacity to separate between rejected points, which may leave more capacity to learn relevant features in order to separate non-rejected points. Another observation is that most of the rejected points were positioned in a central cluster. This structure can make it easier for these points to be captured by the selection function g.

9. Concluding Remarks

We presented SelectiveNet, an effective deep model for selective classification and regression. The power of this new model stems from the mutual training of both the selection and prediction models within the same network, thus forcing the models to focus on the more relevant instances that will not be rejected in production. Known techniques, such as MC-dropout or SR, which typically compose a rejection mechanism over a standard prediction model (trained with full coverage), cannot benefit from this advantage. Our empirical results indicate that the new model currently provides the most accurate classification for a given abstention rate. For selective *regression*, SelectiveNet provides the best and only fast solution, where the only other (inferior) alternative is MC-dropout. This result motivates the use of SelectiveNet in various deep regression applications that require fast inference, such as visual tracking or detection.

We leave a number of issues for future research. As is the case in many deep learning applications, also in selective prediction, ensembling techniques improve performance of basic models. In Lakshminarayanan et al. (2017), it was recently shown that an ensemble of 2-15 models can improve upon the basic SR rejection method. It would be interesting to check if ensembling can improve SelectiveNet as well. Another interesting direction is to examine the influence of capacity (architecture choice) of the selection function g on the overall performance of SelectiveNet. Here, we have not tried to optimize this choice, but we anticipate that different datasets and perhaps different coverage rates may require different architectures also for the selection head.

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